

Mapping coastal aquifer and sea water intrusion using time-domain electromagnetic method, North Oman.

Hesham El-Kaliouby*, Sultan Qaboos University, Earth Science Dept., Oman & National Research Center, Geophysics Dept., Egypt.

Summary

Time Domain Electromagnetic (TDEM) has been widely used in mapping and assessment of groundwater resources and sea water intrusion along coastal alluvial plains. TDEM soundings were used to define aquifer zones in the alluvial aquifer in Al Khoud coastal area in North Oman. The main aquifer zones, composed of clean gravels, clayey gravels, and cemented gravels, could be identified, along with the basal contact of the aquifer. The TDEM data also used to map the depth to the interface between freshwater and saline-water along the coast. The inland extents and thickness of the saline water were mapped along two NW-SE TDEM profiles. The depths to the intrusion zone was investigated using different transmitter loop areas. The resistivity distribution derived from TDEM data agrees closely with the electrical conductivity and resistivity logs from available well data in the study area. TDEM proved to be a successful tool for mapping saline zones and fresh water in arid environment with high depth of penetration and speed of operation.

Introduction

The coastal plain in northern Oman, that extend between Oman mountains and the Gulf of Oman, represent the main source of fresh water. The alluvial coastal aquifer provides an important freshwater resource for both domestic and agricultural use. Almost 50% of the agriculture production is coming from the northern coastal plain, which depends on groundwater as the primary source of agriculture water. However, over pumping from the coastal aquifer resulted in large water-level declines and associated environmental impacts, including groundwater quality deterioration and sea water intrusion, which represents a growing potential threat to water supplies. The salinity increased to levels that affected the crops and human consumption. The yield of the crops, including the date palms which usually tolerate higher salinities, has been affected. Farms have been abandoned as soil salinity has increased. Many wells have been abandoned, all within 6km of the coastline. This highlights the need for urgent conservation measures to sustain this precious economic wealth (Young, et al. 1998, 2004).

The study area (figure 1) is located at Al Khoud fan, northern coastal area of Oman, represent the lower reaches of Samail Catchment, which extends from the mountains region into the coastal plain. Groundwater recharge in the Al Khoud fan occurs from flood runoff originating in the upper Samail catchment. As the volume of abstracted water

has increased, and the water level has declined, sea water has moved landwards. This hazard is highlighted by the shallow depth of the water table on the coastal plain, which is 10-20m from the ground surface. This hazard is highlighted by the shallow depth of the water table, which is less than 1m above mean sea level and often below sea level. For these reasons, there is a great concern and a pressing requirement to assess the available groundwater reserve in the northern coastal plain aquifer (Young, et al. 1998).

It is important to have a fast and reliable tool for regular monitoring of salinity changes along the coastal aquifer in order to get better future prediction and to improve management plans to limit or prevent saltwater intrusion. This is conventionally achieved through drilling a network of monitoring wells along the coastal plane to monitor changes in water salinity over time. However, installation and operation of these wells is expensive and provides limited spatial and vertical coverage. An alternative approach to determine the existence or potential of groundwater resources and saline water intrusion is to use noninvasive geophysical methods (Nenna et.al., 2013).

TDEM is a relatively low cost, rapid, and noninvasive reconnaissance tool to locate potential freshwater resources. TDEM method is well suited to near-surface hydrological studies as it can be used to differentiate between fresh, brackish, and saline water. TDEM method investigates the electrical conductivity of the subsurface through measurements of electromagnetic field over time. The characteristics of the TDEM responses of the earth due to a pulsed primary signal from a large wire square-loop laid on the ground surface yield information about the variations of the electrical conductivity with depth. The conductivity of soils and rocks are controlled by mineralogy, clay content, water content, salinity, and porosity. Changes in the conductivity of soils and rocks, produce variations in the electromagnetic signature. The exact position of the saline water/ fresh-water interface in most coastal areas is poorly constrained on a local level. A combination of monitoring well water quality measurements borehole data, and surface TDEM can be used to accurately map the position of the interface (e.g. Al-Garni and El-Kaliouby, 2011; El-Kaliouby and Abdallah, 2015).

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Field site characteristics

Major groundwater flows in Northern Oman are from the mountains towards the sea. The coastal alluvial deposits in wadi beds form excellent aquifers and contain renewable resources of good quality groundwater by regular recharge from rainfall, surface water flow and locally the inflow of groundwater from adjacent hard rock aquifers. Recharge from local rainfall appears to constitute an important component in the areas, where coarse gravel deposits provide favorable infiltration conditions. Many streams originating in the mountains may flow for an extended period in downstream to recharge coastal aquifer (Stanger, 1986; Young, et al. 2004).

The study area is covered by Quaternary unconsolidated to semi-consolidated sands and gravels with interbedded clay lenses or layers, which rest on clayey conglomerates or directly upon fissured Ophiolites. The Quaternary cover has generally a wedge shape, thickening from a few meters close to the mountains to more than 100 m along the coast. A thick sequence of marls and clays underlies the coarser Quaternary sediments in some areas near the coast, the thickness of this sequence may reaches up to 600 m (Stanger, 1986). Three major units dominate the lithological succession of the alluvium: upper, middle and lower gravel units. The upper gravel unit is composed predominantly of large-size gravels and sand including boulders (figure 2). This unit is loose and very poorly sorted of permeable alluvium with good water quality that makes it the main aquifer zone. The middle unit is discontinuous clayey gravel, which is found in the form of lenses and inter-bedded claystone bands between the upper and lower gravel units. The lower unit is cemented gravely marl and clays, which is more compacted and conglomeratic, due to the cementation caused by diagenesis associated with a fluctuating water table. (Stanger, 1986; Young, et al. 1998).

TDEM Method

TDEM made by transmitting an alternating current into square loops of insulated wire deployed on the land surface. The transmitter loop current consists of equal periods of time-on and time-off base frequencies, which produce an electromagnetic field near the loop of wire. Termination of the current flow is not instantaneous but occurs over a few microseconds, known as the ramp time (or turn-off time), during which the magnetic field is time-variant. The time-variant nature of the induced electromagnetic field creates a primary electromagnetic field in the ground underneath the loop in accordance with Faraday's Law. This resultant field instantly begins to decay, generating additional eddy currents that spread downward and outward into the subsurface like a succession of smoke rings. These eddy currents then produce a secondary magnetic field that is recorded as voltage measurements throughout the time-off

period by a receiver coil positioned in the center or at an off-set from the transmitter loop. The voltage magnitude received from the eddy currents at specific times and depths is determined by the overall conductivity of subsurface rock units and fluids. From these voltage measurements, apparent resistivity values can be calculated. Depth of exploration depends on the transmitter loop dimensions and the injected current intensity (McNeill, 1994).

Data acquisition and processing

A Geonics Protem-47 system (figure 3) using a square transmitter (Tx) loop was used to collect the TDEM soundings. A 1-meter-diameter multi-turn coil receiver (Rx) with a 31.4-square-meter area measured the secondary magnetic fields with central loop and an off-set loop of 25m from the center of the transmitter loop. The Protem-47 transmitter used an injection current of 3.5 amps for the various soundings with preset base frequencies of 25Hz; and 2.5Hz. The Protem TDEM unit samples signal amplitudes in 20 subintervals (time gates) during each measurement and to collect three voltage data sets. Data collected with TDEM method, provided significant contributions to image the extent of saltwater intrusion at many sites in relatively short time. The depths and inland extend of the sea water intrusion was mapped through two profiles perpendicular to the coastline using different loop sizes (20m, and 300m-loop side). The two profiles extend from Al Khoud dam northward towards the sea with a 3 km and 4 km in length (figure 1). The depths of investigations range from tens to hundreds of meters below the ground surface, depending on Tx loop dimensions.

The apparent resistivity data were plotted as a function of time on a log scale. Outliers that deviated severely more than 10 percent above or below other data points were removed or masked before inverse modeling. Inversion is the process of creating an estimate of the true distribution of subsurface resistivity from the measured apparent resistivity obtained from TDEM soundings. Smooth and layered-earth models were generated for each sounding. A smooth model is a vertical delineation of calculated apparent resistivity that represents a non-unique estimate of the true resistivity (Constable, et al., 1987). Ridge regression and Occam's inversion were used by the inversion software (IX1D, Interpex) in a series of iterations to create a smooth model consisting of 20–30 layers for each sounding site. Layered-earth models were then generated based on the results of smooth modeling. The layered-earth models are simplified to fewer layers to represent geologic units with depth. Throughout the area, the layered-earth models range from 3 to 4 layers, depending on observed changes in the raw data and smooth model inversions.

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Interpretation and discussion

The results shown in figures (4 and 5) are for two main NE-SW profiles starting behind Al Khoud dam northward towards the sea. These two profiles (figure 1) were measured by different TDEM loop sizes to be able to penetrate at different depths of investigation. TDEM models show an upper dry and fresh-water saturated gravels followed by an intermediate resistivity that may account to clayey gravel and a bottom low resistivity layer that is attributed to brackish and saline water intrusion. The thickness of the intruded zone increases towards the sea.

Small TDEM loop (20m by 20m) survey covered a depth of 100m showing upper high resistivity layer (100-200 Ohm.m) of dry sand and gravel, that extends to about 15-20m followed by and fresh-water saturated gravels of resistivity (40-90 Ohm.m) and extends to 30m depth. A deeper layer of resistivity (10-40 Ohm.m) that corresponds to clayey gravels extending to a 60-80m depth. A very low resistivity bottom layer (<2 Ohm.m), towards the sea at a depth of more than 80m, is detected and could be attributed to saline interface at the northern stations, since their distance from the sea is about 2km. The small-loop transect showed only the top of the saline zone with a high resolution of the upper layers including the water table but the base of the saline zone is not well defined by this loop dimensions (figure 4).

Larger loop dimensions (300m by 300m) provided a good definition of the base of the saline water at a depth of 250m below ground surface. However, the near surface layers are poorly resolved due to the averaging of resistivities over the area of the loop. This prevent accurate estimation of the resistivity of the overlying aquifer or the other resistivity boundaries of the upper layers that only provide qualitative estimate of the variation of water quality. Below the intrusion zone, an intermediate resistivity layer (20-50 Ohm.m) that could be attributed to cemented gravel composed of low porosity limestone and marl with clay or due to water of lower salinity confined by impermeable clay layer (figure 5).

The results from the two profiles correlates well with water Electrical Conductivity (EC) logs (figure 4), which reflects the total dissolved salt (TDS) within the wells in the study area.

Conclusion

TDEM has proven to be a cost-effective method tool in mapping the fresh/saline water interface and for locating the depth of fresh water aquifer within depths from a few tens of meters to several hundred meters. TDEM offers important advantages over the conventional resistivity depth sounding for mapping the electrically conductive saline zones in the arid environment with improved depth of penetration, depth resolution and speed of acquisition.

The TDEM data collected were considered as good results on the basis of root mean square errors calculated after inversion, comparisons with borehole logging, and repeatability. Models estimated from TDEM data using different transmitter loop sizes suggest that there is a definite geoelectrical boundary between saltwater-saturated and freshwater-saturated sediments, but with different resolutions. Large loop cannot resolve shallow structures in a freshwater-saturated zone, while small loop could resolve it clearly. On the other hand, Large loop data could resolve the saltwater-saturated zone at depth, in which smaller loop could not penetrate to it. Because the different transmitter loops provide complimentary information about the hydrogeological structure at different depths, further analysis using a joint inversion approach of both data sets may improve the estimated geoelectric models of the subsurface. Monitoring or time-lapse TDEM measurements should be carried out at the same time of the year along the same profiles to monitor the movement of the fresh/saline water interface over time.

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References

- Al-Garni, M. El-Kaliouby, H. 2011. Delineation of saline groundwater and sea water intrusion zones using transient electromagnetic (TEM) method, Wadi Thuwal area, Saudi Arabia. *Arabian Journal of Geosciences* 4 (3-4), 655-668
- Constable, S.C., Parker, R.L., and Constable, C.G., 1987. Occam's inversion—A practical algorithm for generating smooth models from EM sounding data: *Geophysics*, 52, p. 289-300.
- El-Kaliouby, H. and Abdallah, O. 2015. Application of time-domain electromagnetic method in mapping saltwater intrusion of a coastal alluvial aquifer, North Oman. *Journal of applied Geophysics* v.115, p 59-64.
- McNeill, J.D., 1994. Principles and application of time domain electromagnetic techniques for resistivity sounding: *Geonics technical note TN-27*, Mississauga, Ontario, p.1-15.
- Nenna, V., Herckenrath, D., Knight, R., Odum, N., and McPhee, D., 2013. Application and evaluation of electromagnetic methods for imaging saltwater intrusion in coastal aquifers: *Seaside Groundwater Basin, California. Geophysics*, 78, p. B77-B88.
- Stanger G., 1986. *The hydrogeology of the Oman Mountains*. PhD Dissertation, Open University, Milton Keynes, England, 355 pp.
- Young, M, Macumber, P , Watts, M and Al Touqy, N. 2004. Electromagnetic detection of freshwater lenses in a hyper-arid terrain. *Journal of Applied Geophysics*, 57, p. 43-61.
- Young, M, de Beuijn, R, and Al-Ismaïly, A. 1998. Exploration of an alluvial aquifer in Oman by time-domain electromagnetic sounding. *Hydrogeology Journal*, v. 6, p. 383-393.

